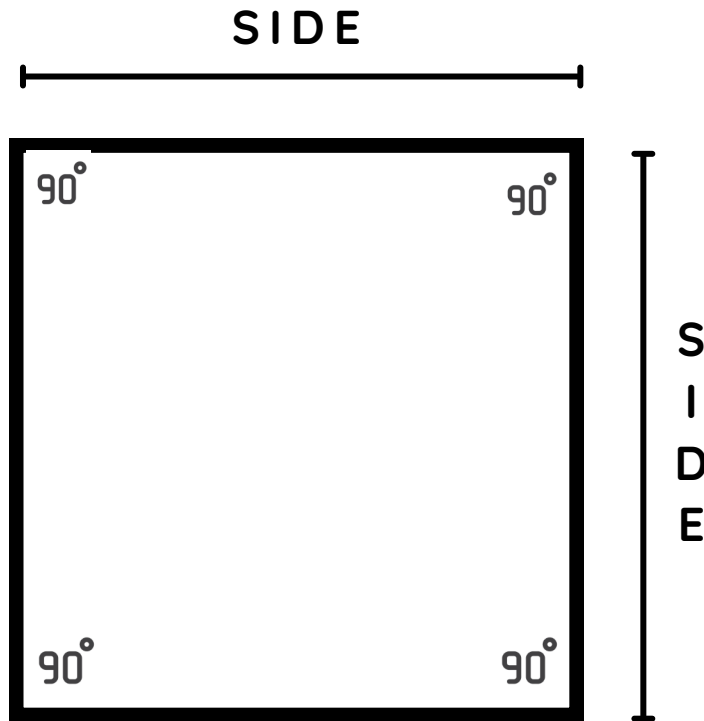


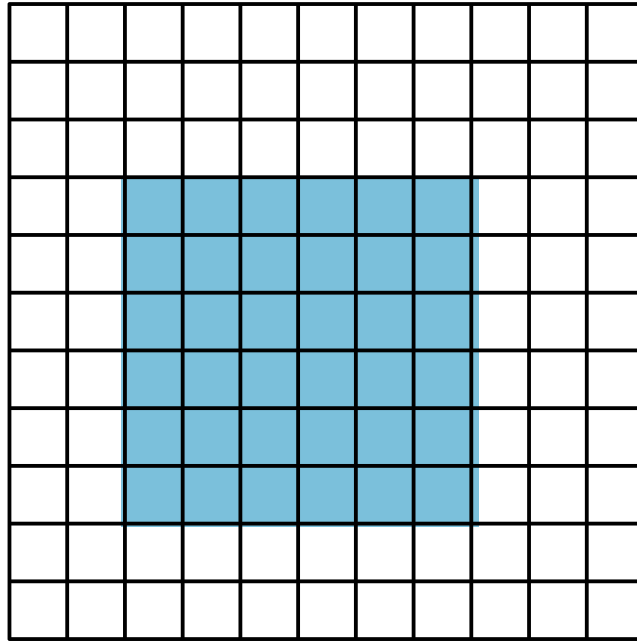
Area and Perimeter of Square

Square



- Has 4 sides.
- Has 4 right angles (each 90 degrees).
- All sides are equal in length.
- Opposite sides are parallel.
- It is a type of quadrilateral, parallelogram, rhombus, and rectangle.
- The diagonals of a square are equal in length, bisect each other, and are perpendicular to each other.

Area on a Grid



A grid has vertical and horizontal lines that form a square. Each square measures 1 sq. unit.

Let's count how many squares the blue rectangle covers.

There are 36 squares.

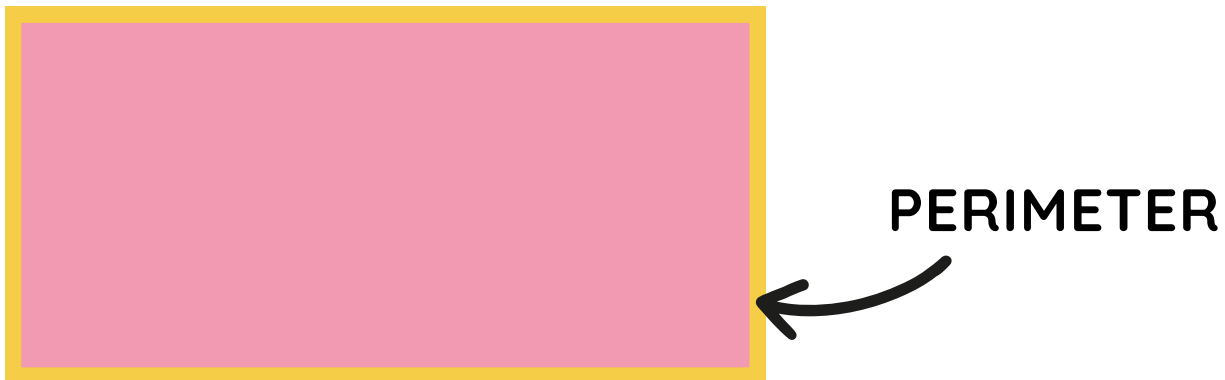
This means that the Square has an area of 36 square units.

Area & Perimeter



Area

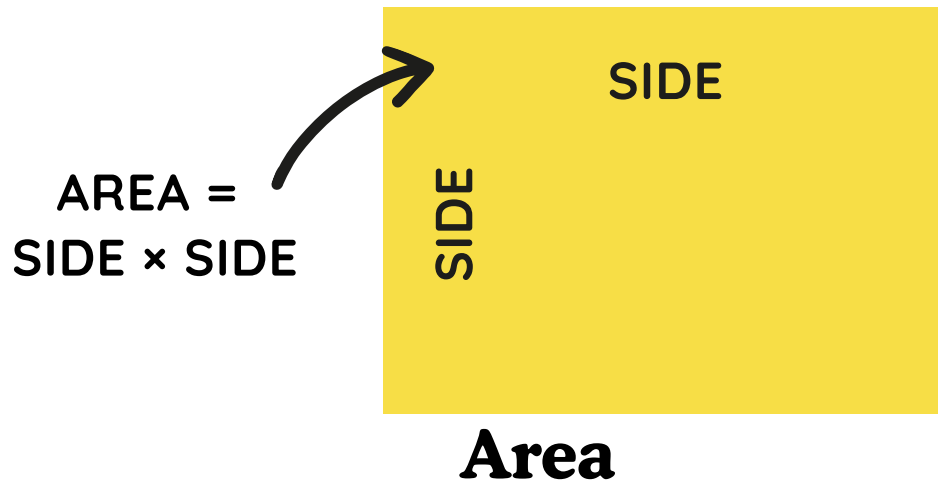
The amount of space inside a shape.



Perimeter

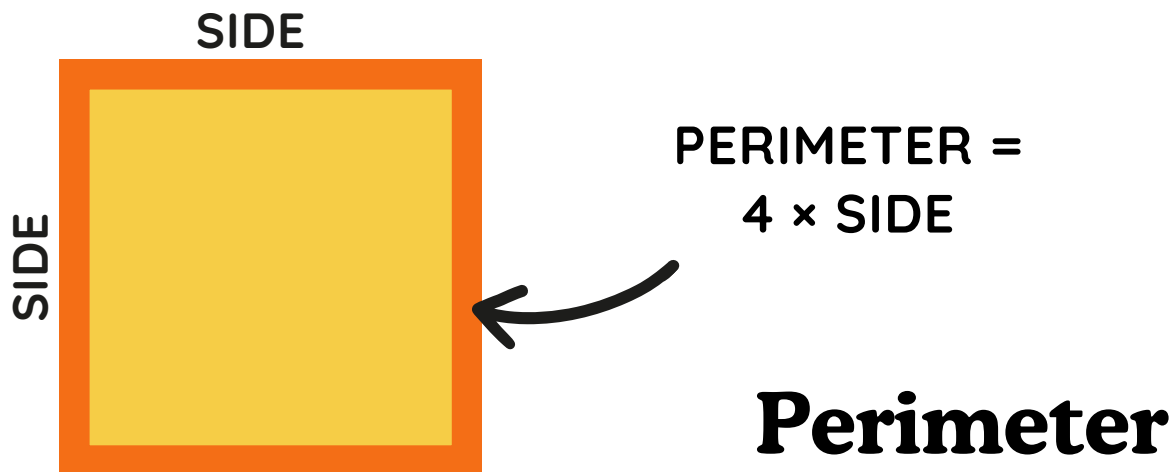
The total distance around the outside of a shape.

Area & Perimeter



The area is the product of a side multiplied by itself, since all sides of a square are equal.

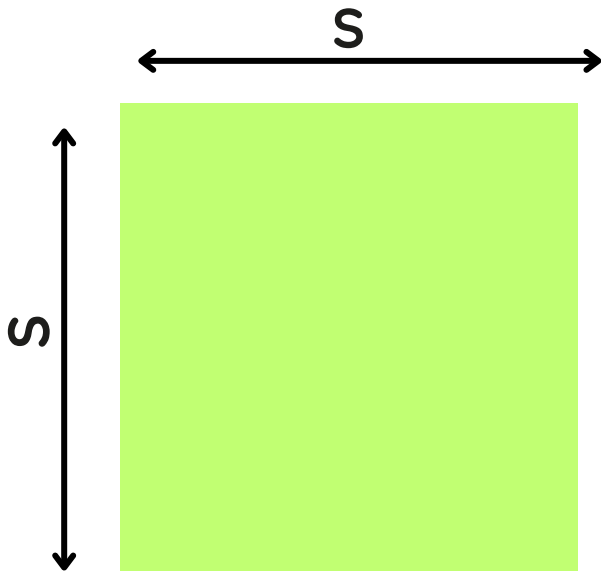
☞ Area is expressed in square units (like cm^2 , m^2 , etc.).



The perimeter is the sum of all four equal sides, which is the same as 4 times one side.

☞ Perimeter is expressed in units (like cm, m, etc.).

Formulas



Area of Square

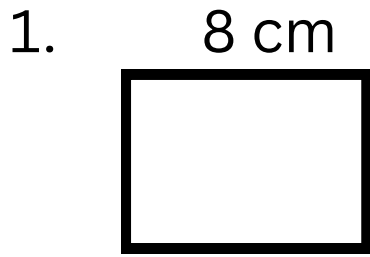
$$A = S \times S$$

Perimeter of Square

$$P = 4S$$

Worksheets

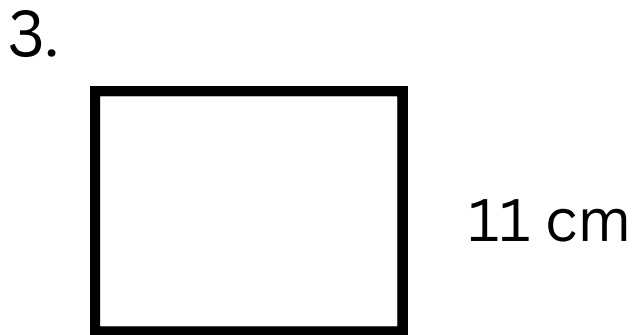
Worksheet 1: Find Area & Perimeter



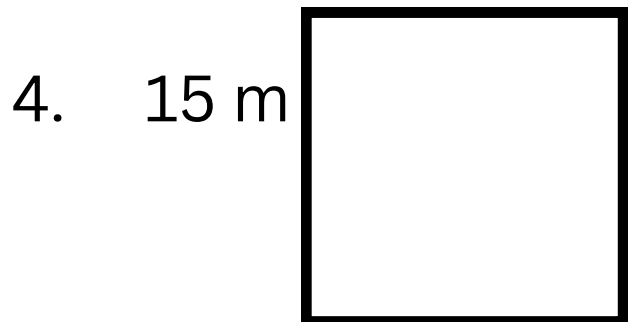
Area = _____ cm^2
Perimeter = _____ cm



Area = _____ m^2
Perimeter = _____ m



Area = _____ cm^2
Perimeter = _____ cm



Area = _____ m^2
Perimeter = _____ m

Worksheet 1: Find Area & Perimeter

(continued)

5.

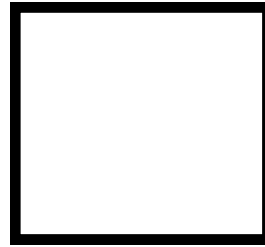


Area = _____ cm^2

Perimeter = _____ cm

6.

12 m



Area = _____ m^2

Perimeter = _____ m

7.

17 cm



Area = _____ cm^2

Perimeter = _____ cm

8.

25 m



Area = _____ m^2

Perimeter = _____ m

Worksheet 2 : Solve the below

1. A square has sides of 6 cm.

◆ Find the area and perimeter.

2. Side of a square = 9 m

◆ Find the area and perimeter.

3. A garden is shaped like a square with each side 15 m long.

◆ What is its area and perimeter?

4. A square has a side length of 10 cm.

◆ Find the area and perimeter.

Worksheet 2 : Solve the below

Continued

5. Find the area and perimeter of a square with:

◆ Side = 12 m

6. A square has an area of 49 cm^2 .

◆ What is the length of one side? Also, find the perimeter.

7. The perimeter of a square is 36 cm.

◆ What is the length of one side? Also, find the area.

8. A floor tile has sides that are 8 cm each.

◆ Is it a square? Find the area and perimeter.

Worksheet 3: Find the side of the Squares

1. The area of a square is 49 cm^2 .

◆ What is the length of one side?

Side = _____ cm

2. The perimeter of a square is 24 cm.

◆ What is the length of one side?

Side = _____ cm

3. A square has an area of 64 m^2 .

◆ What is the side length?

Side = _____ m

4. A square has a perimeter of 36 cm.

◆ What is the side length?

Side = _____ cm

Worksheet 3: Find the side of the Squares

continued

5. The area of a square is 100 m^2 .

◆ What is the length of one side?

Side = _____ m

6. A square has a perimeter of 48 cm.

◆ Find the side length.

Side = _____ cm

7. The area of a square is 81 cm^2 .

◆ What is the side length?

Side = _____ cm

8. A square has a perimeter of 60 m.

◆ Find the side length.

Side = _____ m

Worksheet 4: Word Problems

1. Riya wants to decorate a square table with a cloth that covers the whole top. If each side of the table is 5 meters,

- ◆ What is the area of the table?
- ◆ What is the perimeter of the table?

2. A square-shaped photo frame has an area of 81 cm^2 .

- ◆ What is the length of one side of the frame?
- ◆ What is the perimeter?

3. A playground is shaped like a square. The distance around it (perimeter) is 64 meters.

- ◆ What is the length of each side?

4. A square tile has sides of 8 cm.

- ◆ What is its area?
- ◆ How much border would be needed to go all around one tile?

Worksheet 4: Word Problems

Continued

5. An artist is painting a square canvas that has an area of 100 m^2 .

- ◆ What is the side length of the canvas?
- ◆ What is its perimeter?

6. A square garden has a perimeter of 36 meters.

- ◆ What is the length of each side?
- ◆ What is the area of the garden?

7. Sara cut a square-shaped paper with side 9 cm.

- ◆ Find the area and perimeter of the paper.

8. The area of a square field is 196 m^2 .

- ◆ What is the length of one side?
- ◆ How many meters of fencing is needed to go all around the field (perimeter)?

Worksheet 1 : Find Area and Perimeter

Answers

1.Side = 8 cm

✎ Area = $8 \times 8 = 64 \text{ cm}^2$

✎ Perimeter = $4 \times 8 = 32 \text{ cm}$

2.Side = 7 m

✎ Area = $7 \times 7 = 49 \text{ m}^2$

✎ Perimeter = $4 \times 7 = 28 \text{ m}$

3.Side = 11 cm

✎ Area = $11 \times 11 = 121 \text{ cm}^2$

✎ Perimeter = $4 \times 11 = 44 \text{ cm}$

4.Side = 15 m

✎ Area = $15 \times 15 = 225 \text{ m}^2$

✎ Perimeter = $4 \times 15 = 60 \text{ m}$

5.Side = 4 cm

✎ Area = $4 \times 4 = 16 \text{ cm}^2$

✎ Perimeter = $4 \times 4 = 16 \text{ cm}$

6.Side = 12 m

✎ Area = $12 \times 12 = 144 \text{ m}^2$

✎ Perimeter = $4 \times 12 = 48 \text{ m}$

7.Side = 17 cm

✎ Area = $17 \times 17 = 289 \text{ cm}^2$

✎ Perimeter = $4 \times 17 = 68 \text{ cm}$

8.Side = 25 m

✎ Area = $25 \times 25 = 625 \text{ m}^2$

✎ Perimeter = $4 \times 25 = 100 \text{ m}$

Worksheet 2 : Solve the below

Answers

1.Side = 6 cm

✎ Area = $6 \times 6 = 36 \text{ cm}^2$

✎ Perimeter = $4 \times 6 = 24 \text{ cm}$

2.Side = 9 m

✎ Area = $9 \times 9 = 81 \text{ m}^2$

✎ Perimeter = $4 \times 9 = 36 \text{ m}$

3.Side = 15 m

✎ Area = $15 \times 15 = 225 \text{ m}^2$

✎ Perimeter = $4 \times 15 = 60 \text{ m}$

4Side = 10 cm

✎ Area = $10 \times 10 = 100 \text{ cm}^2$

✎ Perimeter = $4 \times 10 = 40 \text{ cm}$

5.Side = 12 m

✎ Area = $12 \times 12 = 144 \text{ m}^2$

✎ Perimeter = $4 \times 12 = 48 \text{ m}$

6.Area = 49 cm^2

✎ Side = $\sqrt{49} = 7 \text{ cm}$

✎ Perimeter = $4 \times 7 = 28 \text{ cm}$

7.Perimeter = 36 cm

✎ Side = $36 \div 4 = 9 \text{ cm}$

✎ Area = $9 \times 9 = 81 \text{ cm}^2$

8.Side = 8 cm

✎ Yes, it is a square (all sides equal)

✎ Area = $8 \times 8 = 64 \text{ cm}^2$

✎ Perimeter = $4 \times 8 = 32 \text{ cm}$

Worksheet 3: Find the side of the Squares

Answers

1. Area = 49 cm^2

👉 Side = $\sqrt{49} = 7 \text{ cm}$

2. Perimeter = 24 cm

👉 Side = $24 \div 4 = 6 \text{ cm}$

3. Area = 64 m^2

👉 Side = $\sqrt{64} = 8 \text{ m}$

4. Perimeter = 36 cm

👉 Side = $36 \div 4 = 9 \text{ cm}$

5. Area = 100 m^2

👉 Side = $\sqrt{100} = 10 \text{ m}$

6. Perimeter = 48 cm

👉 Side = $48 \div 4 = 12 \text{ cm}$

7. Area = 81 cm^2

👉 Side = $\sqrt{81} = 9 \text{ cm}$

8. Perimeter = 60 m

👉 Side = $60 \div 4 = 15 \text{ m}$

Worksheet 4: Word Problems

Answers

1.Side = 5 m

☞ Area = side $\times$ side = $5 \times 5 = 25 \text{ m}^2$

☞ Perimeter = $4 \times \text{side} = 4 \times 5 = 20 \text{ m}$

2.Area = 81 cm^2

☞ Side = $\sqrt{81} = 9 \text{ cm}$

☞ Perimeter = $4 \times 9 = 36 \text{ cm}$

3.Perimeter = 64 m

☞ Side = $64 \div 4 = 16 \text{ m}$

4.Side = 8 cm

☞ Area = $8 \times 8 = 64 \text{ cm}^2$

☞ Perimeter = $4 \times 8 = 32 \text{ cm}$

5.Area = 100 m^2

☞ Side = $\sqrt{100} = 10 \text{ m}$

☞ Perimeter = $4 \times 10 = 40 \text{ m}$

6.Perimeter = 36 m

☞ Side = $36 \div 4 = 9 \text{ m}$

☞ Area = $9 \times 9 = 81 \text{ m}^2$

7.Side = 9 cm

☞ Area = $9 \times 9 = 81 \text{ cm}^2$

☞ Perimeter = $4 \times 9 = 36 \text{ cm}$

8.Area = 196 m^2

☞ Side = $\sqrt{196} = 14 \text{ m}$

☞ Perimeter = $4 \times 14 = 56 \text{ m}$